

Model Architecture & Training Hyperparameters

Component	Parameter	Value	Rationale
Embedding	Input -> d_model	V -> 128 -> 64	Two-layer MLP + GELU
	Positional Encoding	Learnable (2000, 64)	Adapts to variable-length input
Liquid-Gated SSM x 2	d_model	64	Balanced capacity vs overfitting
	d_state (SSM state dim)	16	Small state for sample efficiency
	d_conv (causal kernel)	4	Captures hourly local patterns
	Expand ratio	2x (d_inner=128)	Standard Mamba-2 expansion
LNN Dynamic Gate	Backbone	GRU (1 layer)	ODE discretization $\approx$ liquid TC
	Hidden dim (GEFCom2014)	32	Lightweight: only 1.7% of total params
	Hidden dim (GEFCom2012)	48	Larger for richer NWP features
	Gate outputsigma(W*h_GRU) in Per-channel	Per-channel	Per-channel, per-timestep modulation
Quantile Decoder	Architecture	d -> 2d -> d -> 24x99	Three-layer + GELU + Dropout(0.1)
	Output shape	(B, 24, 99)	Joint 99-quantile, all horizons
Training	Optimizer	AdamW (lr=1e-3, wd=1e-5)	Standard for time series DL
	LR Scheduler	CosineAnnealingWarmRestarts	T0=15, T_mult=2
	Batch Size	48–64	GPU-memory constrained
	Precision	AMP (float16 + GradScaler)	2x speedup, stable training
	Gradient Clipping	max_norm = 1.0	Prevents explosion in long sequences
	Early Stopping	Patience = 8–12 epochs	On validation pinball loss
Total Parameters		~412,000	Single consumer GPU (8GB VRAM)
Inference Speed		~5 ms / window	Real-time deployment feasible

All variants share core architecture (d=64, ds=16, nb=2). Only LNN hidden dim and training data vary across experiments.